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Student number



2024 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Student Number at the top of this page

Total marks:
100

Section I – 10 marks (pages 2–7)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8–32)

- Attempt Questions 11–35
- Allow about 2 hours and 45 minutes for this section

Section I

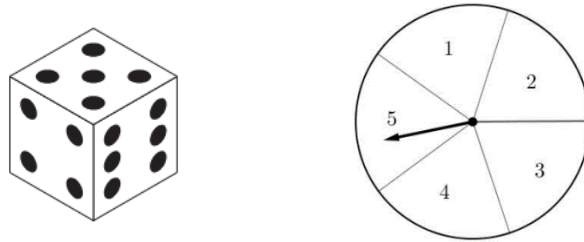
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 A game involves throwing a standard die and spinning a spinner numbered 1 to 5.



The product of the two numbers is the score.

The table of scores below is partially completed.

		SPINNER				
		1	2	3	4	5
DIE	1	1	2	3	4	
	2	2	4	6		
	3	3	6			
	4	4				
	5					
	6					

What is the probability of obtaining an even score?

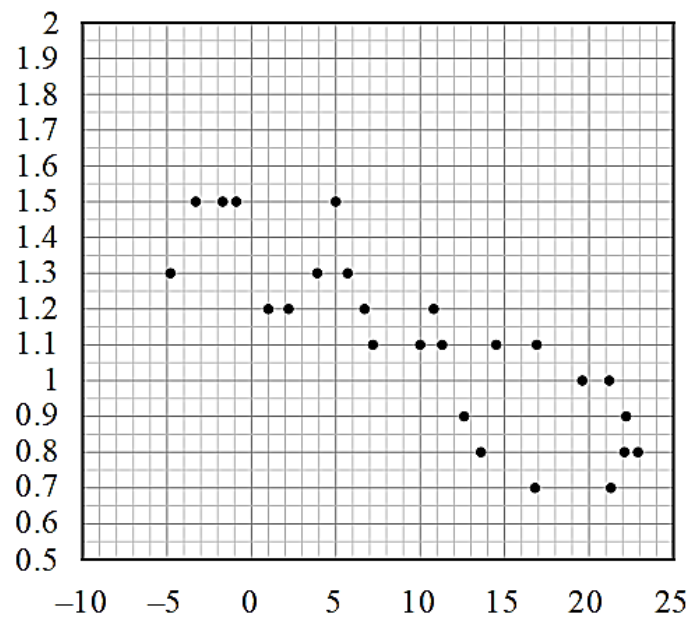
- A. $\frac{3}{10}$
- B. $\frac{5}{10}$
- C. $\frac{6}{10}$
- D. $\frac{7}{10}$

- 2 An athlete runs for 500 metres around a circular track with radius 200 metres.
How far, in a direct line across the field, is the athlete from her starting position?

A. 200 m
B. 283 m
C. 380 m
D. 458 m

- 3 A researcher assumes that there is a linear relationship between the temperature and hot chocolate consumption.

Data was collected randomly in cities across Europe and the results were recorded in the scatterplot below.



The scatterplot shows the average monthly hot chocolate consumption, in litres/person against average monthly temperature, in °C.

The correlation coefficient for this data would be closest to:

A. -0.8
B. -0.4
C. 0.4
D. 0.8

- 4 A set of scores in a data set has a median of 72, an interquartile range of 14 and a variance of 4.

Each score has 3 added to it and each resulting score is then doubled to form a new set of scores.

What is the median, interquartile range, and variance of the new set of scores?

	Median	Interquartile range	Variance
A.	144	28	8
B.	150	28	16
C.	150	32	8
D.	150	32	16

- 5 How many solutions does $\sec \theta + \tan \theta = 0$ have for $-\pi \leq \theta \leq \pi$?

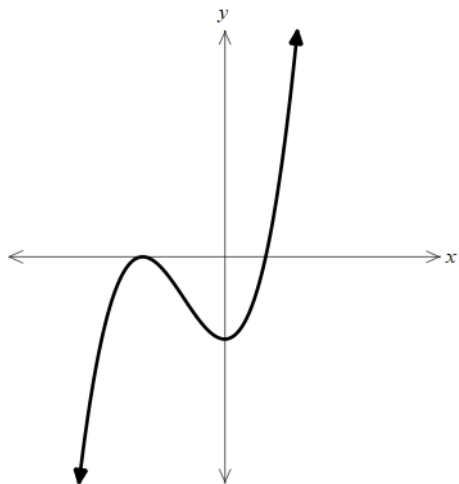
- A. 0
- B. 1
- C. 2
- D. 3

- 6 Let $g(x) = \sqrt{2-x}$. What is the domain of the composite function $g(g(x))$?

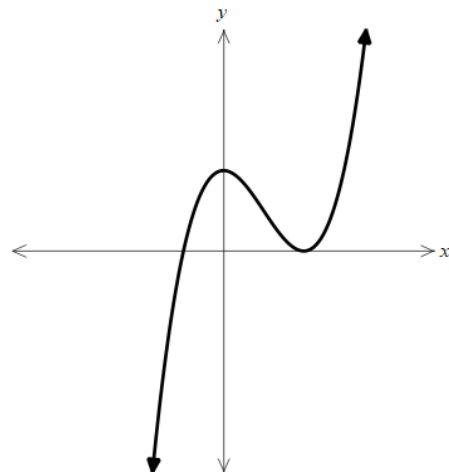
- A. $x \leq \sqrt{2}$
- B. $x \leq 2$
- C. $-\sqrt{2} \leq x \leq \sqrt{2}$
- D. $-2 \leq x \leq 2$

7 Which of the following shows the graph of $f(x) = (x+a)^2(b-x)$, if $a > 0$ and $b > 0$?

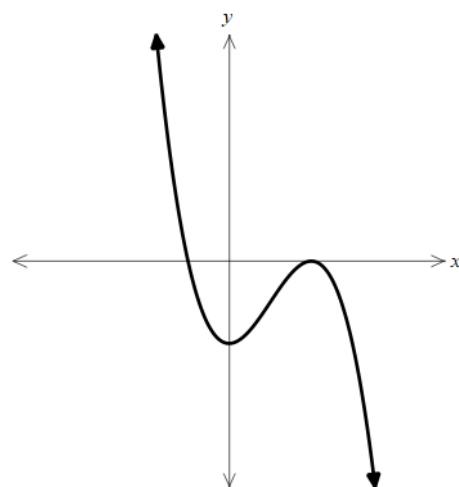
A.



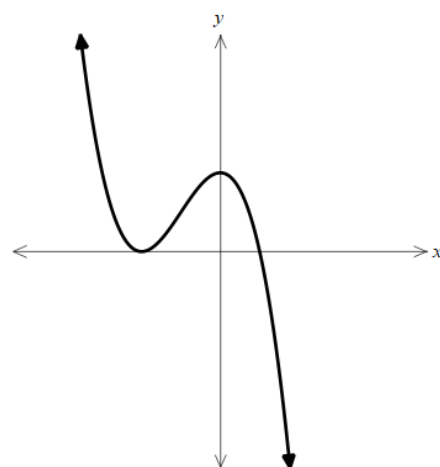
B.



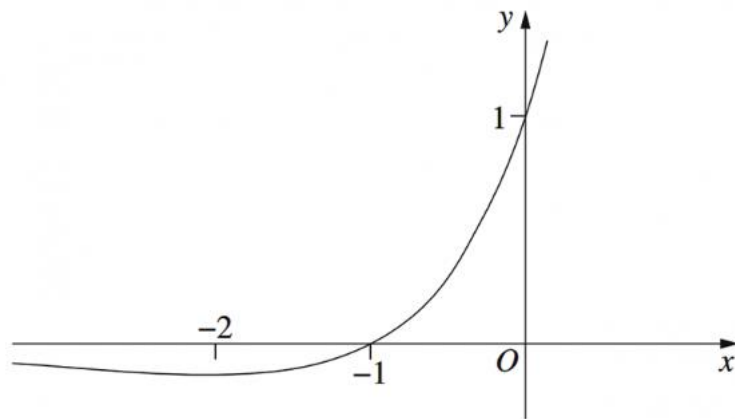
C.



D.



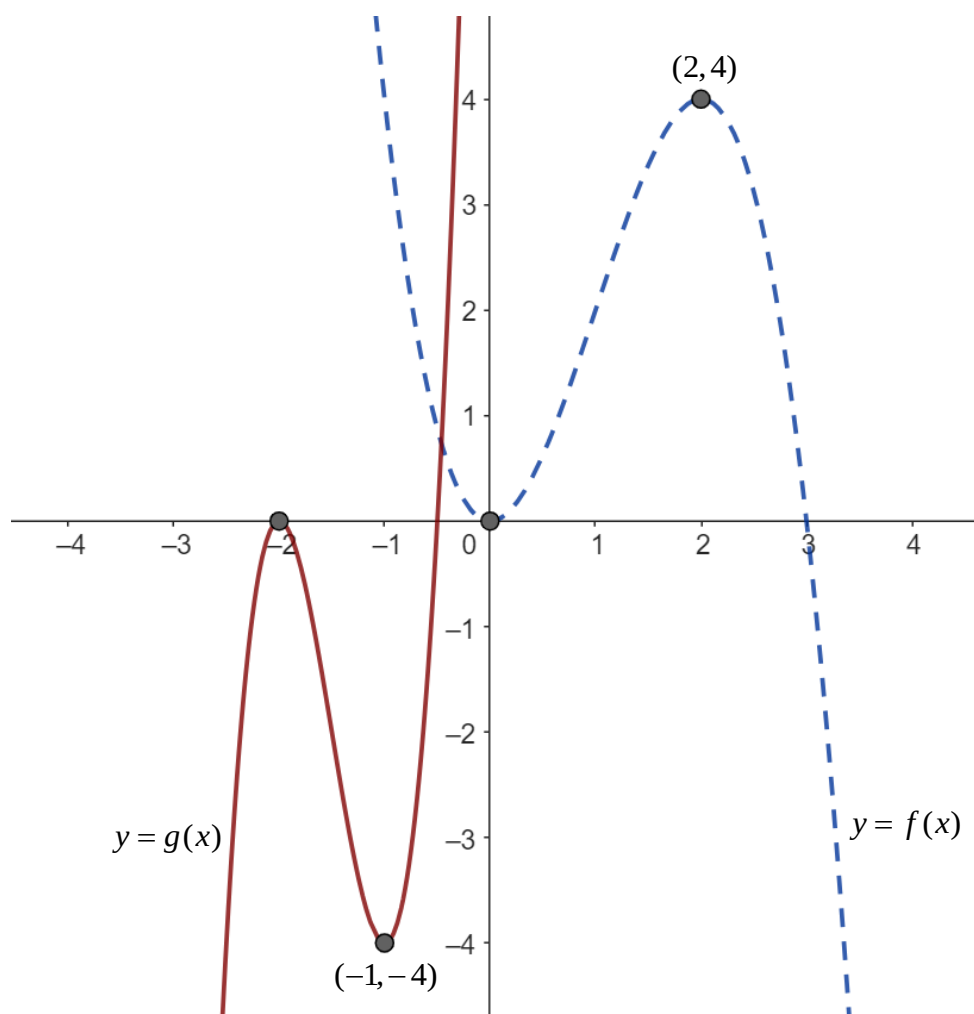
- 8 The diagram shows the graph of $y = e^x(1+x)$.



How many solutions are there to the equation $e^x(1+x) = 1 - (x+1)^2$?

- A. 0
B. 1
C. 2
D. 3
- 9 Given that $\int_1^6 f(x) dx = -2$ and $\int_3^6 f(x) dx = 6$, what is the value of $\int_1^3 (f(x) + x) dx$?
- A. -8
B. -4
C. 4
D. 8

- 10 The graphs of $y = f(x)$ and $y = g(x)$ are shown. The function $y = g(x)$ can be obtained by applying a series of transformations to $y = f(x)$.



Which of the following is correct?

- A. $g(x) = -f(x+4)$
- B. $g(x) = -f(2x+4)$
- C. $g(x) = f(-2x) - 4$
- D. $g(x) = f(-x+2) - 4$

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Student number



Mathematics Advanced Section II Answer booklet A

50 marks

Attempt Questions 11–26

Allow about 1 hour and 30 minutes for this section

Instructions

- Write your Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 11 (2 marks)

ABC is a triangle in which $\cos A = -\frac{2}{5}$. Find the exact value of $\tan A$.

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Question 12 (2 marks)

Solve $|3x + 2| = 4$

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Question 13 (3 marks)

Find the exact equation of the tangent to the curve $y = \log_2 x$ at the point where $x = 4$.

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Question 14 (4 marks)

Barry owns an ice cream cart. The table below shows the number of ice creams Barry sold in a one hour period, on eight different days whose temperatures were measured in degrees Celsius.

Temperature	18	21	23	26	27	29	32	34
Ice creams sold	2	4	9	16	16	25	34	37

- a) Find the equation of the least-squares line of best fit for this data, rounding your answers to 2 decimal places. **2**

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- b) Use your equation to predict how many ice creams Barry would sell in one hour when the temperature is 30°C. **1**

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- c) Explain why the value of the y intercept for the line of best fit has no meaning in this context. **1**

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Question 15 (2 marks)

Differentiate $\frac{(2x+1)^2}{4+x}$ with respect to x

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Question 16 (3 marks)

a) Show that $\frac{d}{dx}(x^2 e^{x^2}) = 2xe^{x^2}(1+x^2)$.

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b) Hence show that $y = x^2 e^{x^2}$ is decreasing for all $x < 0$.

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Question 17 (2 marks)

Find $\int \frac{x-3}{x^2-6x} dx$

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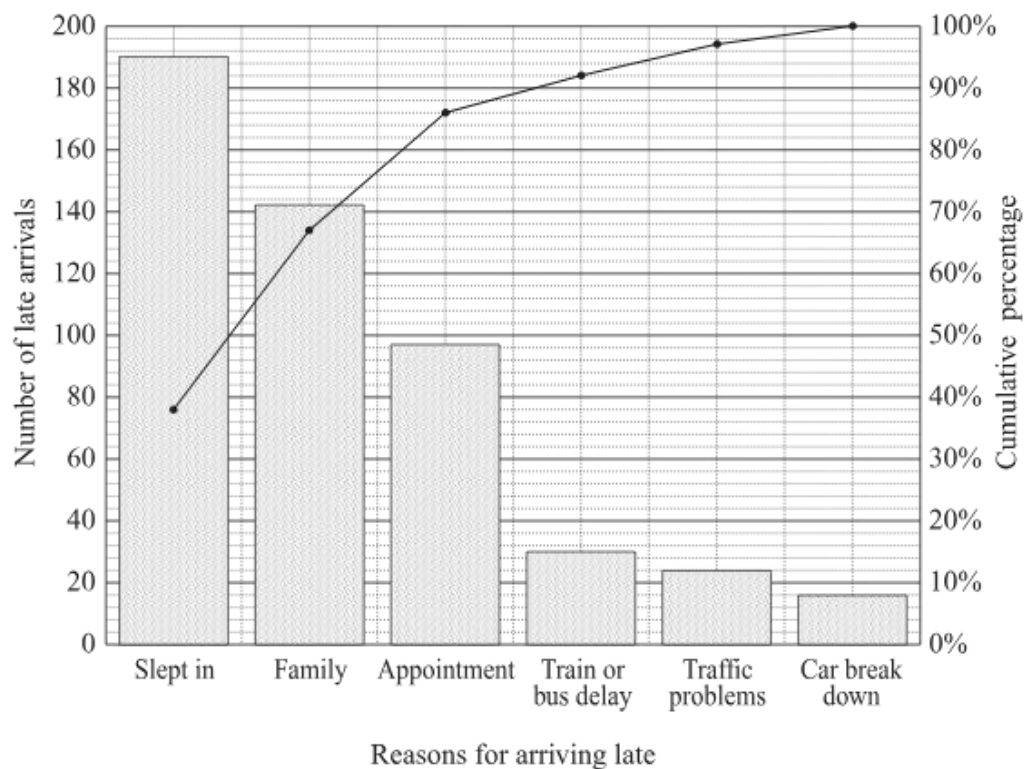
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Question 18 (1 mark)

A school collected data on the reasons given by students for arriving late. The data is displayed in the Pareto chart below.

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What percentage of students gave the reasons traffic problems or car break down?

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Question 19 (4 marks)

A bag of potatoes is labelled as having a net weight of 2 kilograms. The packaging process produces bags of potatoes whose weights are normally distributed with a mean of 2.05 kilograms and a standard deviation of 0.05 kilograms.

- a) In a supermarket selling 2000 of these bags of potatoes, how many bags will contain less than the labelled weight? 2

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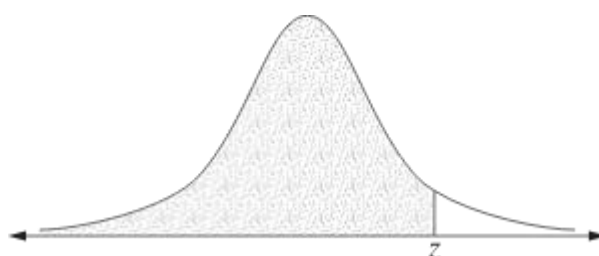
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- b) For a random variable which is normally distributed with a mean of 0 and a standard deviation of 1, the table gives the probability that this random variable lies below z for some positive values of z . 2

z	0.80	0.81	0.82	0.83	0.84	0.85	0.86	0.87
Probability	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078

The probability values given in the table are represented by the shaded area in the following diagram.



Using this table, above what weight would the heaviest 20% of bags of potatoes weigh?

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Question 20 (5 marks)

Sixty people visited a swimming pool one afternoon. Thirteen out of the twenty-three people who wore goggles were adults. There were nineteen children.

- a) Complete the two-way table below, showing this information.

2

	Wore goggles	No goggles	Total
Adults			
Children			
Total			

- b) If an adult is selected at random, what is the probability they were wearing goggles?

1

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- c) An additional group of adults, not wearing goggles, arrived at the pool.
The probability that a person selected at random is an adult who is not wearing goggles is now 0.6.

2

How many additional adults arrived?

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Question 21 (8 marks)

The derivative of a function $y = f(x)$ is given by $f'(x) = 3x^2 - 4x + 1$

- a) Find the x -coordinates of the stationary points of $y = f(x)$ and determine the nature of the stationary points. **3**

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- b) The curve $y = f(x)$ has a y -intercept at $y = -2$. **2**

Find an expression for $f(x)$.

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- c) Sketch the curve, clearly indicating the stationary points and intercepts on the axes. 2

- d) For what values of x is the curve concave down? 1

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Question 22 (3 marks)

A and B are two events such that $P(A) = 0.3$, $P(B) = 0.5$ and $P(A|B) = 0.4$.

a) Show that $P(A \cup B) = 0.6$

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b) Find $P(\text{neither } A \text{ nor } B)$

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Question 23 (3 marks)

Find two possible equations of a semi-circle with range $[3, 5]$, given that it is an even function.

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Question 24 (2 marks)

A lift accelerates from rest at a constant rate until it reaches a speed of 5 metres per second. It continues at this speed for 12 seconds and then decelerates at a constant rate before coming to rest. The total travel time for the lift is 30 seconds.

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Find the total distance, in metres, travelled by the lift.

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Question 25 (3 marks)

The table shows the probability distribution of a discrete random variable.

x	0	1	2	3	4
$P(X = x)$	$p^2 + p$	p^2	p	$2p^2$	p

- a) Show that $p = \frac{1}{4}$.

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- b) Find the expected value $E(X)$

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Question 26 (3 marks)

Show that $\log_e(1 + \sin x) - \log_e(1 - \sin x) = 2\log_e(\sec x + \tan x)$

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Student number



Mathematics Advanced Section II Answer booklet B

40 marks

Attempt Questions 27–35

Allow about 1 hour and 15 minutes for this section

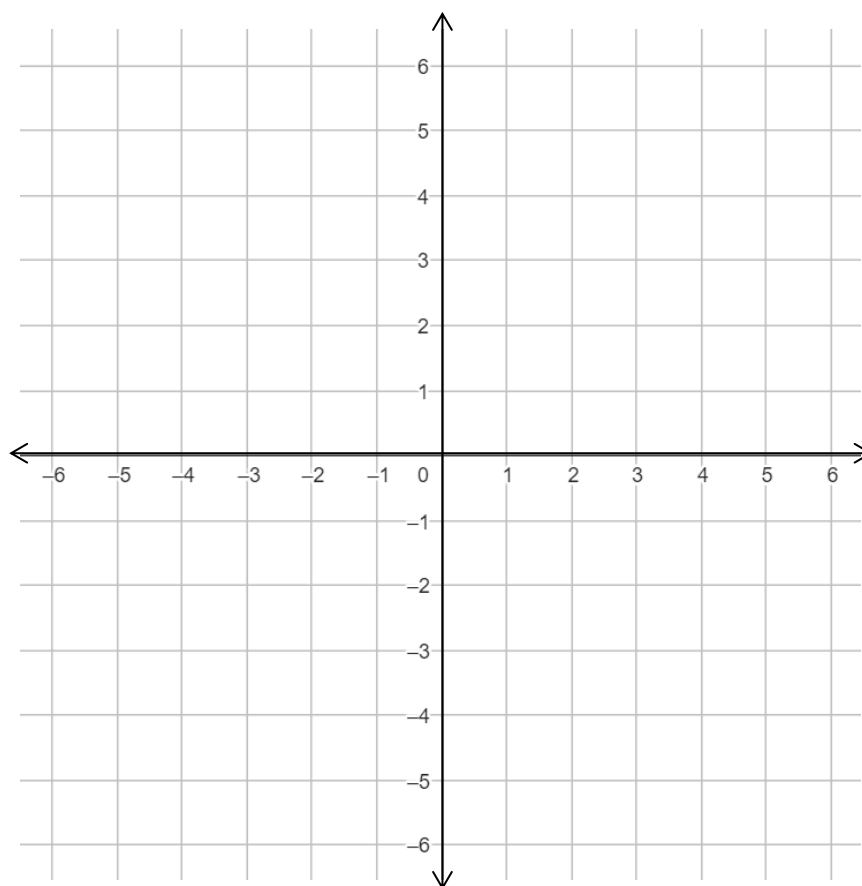
Instructions

- Write your Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 27 (6 marks)

- a) Sketch the graphs of the functions $y = \frac{4}{x}$ and $y = |x+1| - 4$ on the number plane below. Clearly show their point(s) of intersection and any intercepts on the axes.

4



- b) Hence, or otherwise, solve $|x+1| - 4 \leq \frac{4}{x}$

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Question 28 (5 marks)

Consider $f(x) = \sqrt{1-x^2}$

- a) Use the trapezoidal rule with 4 subintervals to estimate $\int_0^1 \sqrt{1-x^2} dx$. **2**

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- b) Find the exact value of $\int_0^1 \sqrt{1-x^2} dx$ and hence find an approximation for π correct to 3 decimal places. **2**

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- c) Explain why this value under-estimates the exact value of π . **1**

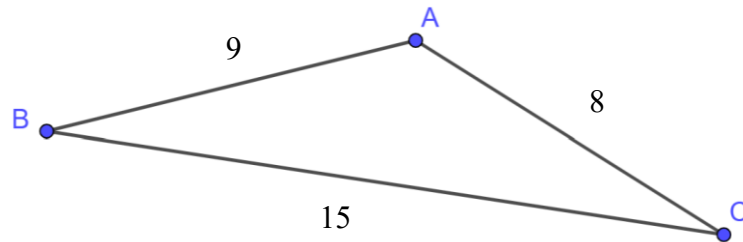
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Question 29 (4 marks)

The diagram shows a triangle ABC where $AB = 9$, $AC = 8$ and $BC = 15$.

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Find the area of the triangle ABC . Give your answer correct to one decimal place.

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Question 30 (4 marks)

a) Show that $\frac{\operatorname{cosec} x}{\operatorname{cosec} x - \sin x} \equiv \sec^2 x$.

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b) Hence, or otherwise, find the exact value of $\int_0^{\frac{\pi}{3}} \tan^2 x \left(\frac{\operatorname{cosec} x}{\operatorname{cosec} x - \sin x} \right) dx$

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Question 31 (1 mark)

For a given discrete probability distribution, $E(X) = 5$ and $\text{Var}(X) = 3$.

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Find the value of $E(2X + 1)$.

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Question 32 (5 marks)

A wire of length L is bent to form a sector with radius r and angle θ .

a) Show that $\theta = \frac{L - 2r}{r}$.

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b) Show that the area A of the sector is maximised when $r = \frac{L}{4}$.

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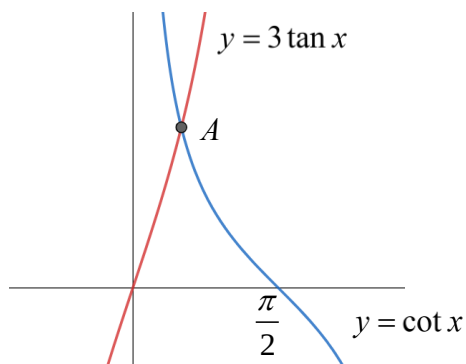
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Question 33 (5 marks)

Part of each of the curves $y = 3 \tan x$ and $y = \cot x$ are shown on the graph below.

The curve $y = \cot x$ cuts the x axis at $x = \frac{\pi}{2}$ and the two curves intersect at the point A .



- a) Show that the x coordinate of A is $x = \frac{\pi}{6}$.

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- b) By considering $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$, find the exact area enclosed by the curves and the x axis between $x = 0$ and $x = \frac{\pi}{2}$.

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Question 34 (2 marks)

A farming company harvests oranges to sell to grocery stores. Oranges that weigh less than 150 grams are rejected and used for juicing instead. Typically, 16% of all oranges are rejected. Also, oranges that are larger than 300 grams are reserved for restaurants. Typically, 2.5% of all oranges are sold to restaurants.

2

If the weights of the oranges are normally distributed, find the mean and standard deviation of the normal distribution.

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Question 35 (8 marks)

A continuous random variable X has the probability density function $f(x)$ given by

$$f(x) = \begin{cases} k(\sin(\pi x) + 2x), & \text{for } 0 \leq x \leq 3 \\ 0, & \text{for all other values of } x \end{cases}$$

- a) Show that $k = \frac{\pi}{2 + 9\pi}$.

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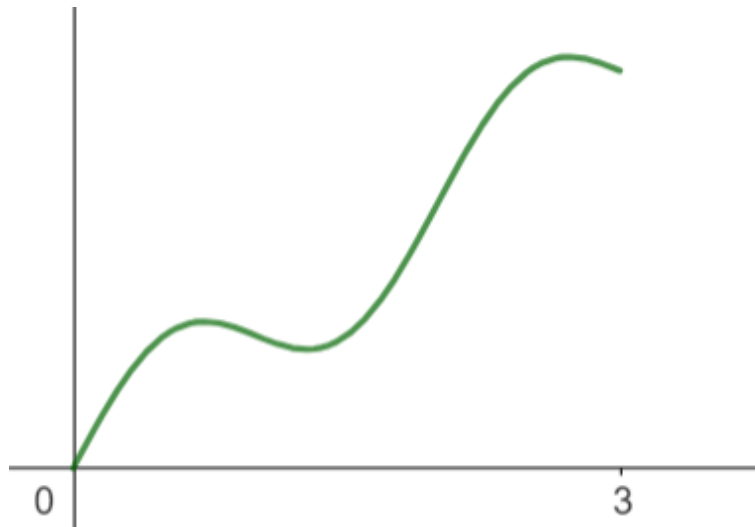
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b) The graph of the density function is given below.

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Without any calculation, explain why the mode must be greater than the median.

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c) Find the mode of X . Give your answer correct to two decimal places.

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End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Student number



Mathematics Advanced

Section I Multiple choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

Section I Student responses

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |